



ENGINEERING MATHEMATICS - II

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ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 8

Sub Topic : Problems on Practical Harmonic Analysis

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Suppose we have a set of N values of $y = f(x)$ having period 2π or $2l$ at equidistant points of x in the interval $(0, 2\pi)$ or $(0, 2l)$ then Fourier coefficients a_0 , a_n and b_n are calculated using the relations given below:

Period	Fourier Coefficients		
2π	$a_0 = \frac{2}{N} \sum y$	$a_n = \frac{2}{N} \sum y \cos nx$	$b_n = \frac{2}{N} \sum y \sin nx$
$2l$	$a_0 = \frac{2}{N} \sum y$	$a_n = \frac{2}{N} \sum y \cos\left(\frac{n\pi x}{l}\right)$	$b_n = \frac{2}{N} \sum y \sin\left(\frac{n\pi x}{l}\right)$

Note:

1. Here **N** denotes the **number of observations**.
2. The amplitude of the first harmonic is given by $\sqrt{a_1^2 + b_1^2}$.
Similarly the amplitudes of the second and third harmonics are given by $\sqrt{a_2^2 + b_2^2}$ and $\sqrt{a_3^2 + b_3^2}$ respectively.

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1. Determine the constant term and the first two coefficients of cosine and sine terms in the Fourier series for the following data.

x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$
y	0	9.2	14.4	17.8	17.3	11.7

Solution:

The interval of x is $0 \leq x < \pi$.

Here, length of interval is π . That is, $l = \frac{\pi}{2}$

Here $N=6$.

$$f(x) = \frac{a_0}{2} + (a_1 \cos \frac{\pi x}{l} + b_1 \sin \frac{\pi x}{l}) + (a_2 \cos \frac{2\pi x}{l} + b_2 \sin \frac{2\pi x}{l}) + \dots$$

Substituting for $l = \frac{\pi}{2}$

$$f(x) = \frac{a_0}{2} + (a_1 \cos 2x + b_1 \sin 2x) + (a_2 \cos 4x + b_2 \sin 4x) + \dots$$

$$\begin{aligned}a_0 &= \frac{2}{N} \sum y \\ &= \frac{2}{6} \sum y = \frac{1}{3} (70.4) = 23.46\end{aligned}$$

$$a_n = \frac{2}{N} \sum y \cos \left(\frac{n\pi x}{l} \right)$$

$$= \frac{2}{6} \sum y \cos \left(\frac{n\pi x}{\pi/2} \right)$$

$$a_n = \frac{1}{3} \sum y \cos (2nx)$$

Put $n=1,2$

$$a_1 = \frac{1}{3} \sum y \cos 2x$$

$$a_2 = \frac{1}{3} \sum y \cos 4x$$

$$b_n = \frac{2}{N} \sum y \sin \left(\frac{n\pi x}{l} \right)$$

$$= \frac{2}{6} \sum y \sin \left(\frac{n\pi x}{\pi/2} \right)$$

$$b_n = \frac{1}{3} \sum y \sin (2nx)$$

$$b_1 = \frac{1}{3} \sum y \sin 2x$$

$$b_2 = \frac{1}{3} \sum y \sin 4x$$

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x	y	$\cos 2x$	$y \cos 2x$	$\cos 4x$	$y \cos 4x$
0	0	1	0	1	0
$\pi/6$	9.2	0.5	4.6	-0.5	-4.6
$\pi/3$	14.4	-0.5	-7.2	-0.5	-7.2
$\pi/2$	17.8	-1	-17.8	1	17.8
$2\pi/3$	17.3	-0.5	-8.65	-0.5	-8.65
$5\pi/6$	11.7	0.5	5.85	-0.5	-5.85
Total			-23.2		-8.5

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$$\begin{aligned} a_1 &= \frac{1}{3} \sum y \cos 2x \\ &= \frac{1}{3} [-23.2] = -7.73 \end{aligned}$$

$$\begin{aligned} a_2 &= \frac{1}{3} \sum y \cos 4x \\ &= \frac{1}{3} [-8.5] = -2.83 \end{aligned}$$

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x	y	$\sin 2x$	$y \sin 2x$	$\sin 4x$	$y \sin 4x$
0	0	0	0	0	0
$\pi/6$	9.2	0.866	7.9672	0.866	7.9672
$\pi/3$	14.4	0.866	12.4704	-0.866	-12.4704
$\pi/2$	17.8	0	0	0	0
$2\pi/3$	17.3	-0.866	-14.9818	0.866	14.9818
$5\pi/6$	11.7	-0.866	-10.1322	-0.866	-10.1322
Total			-4.6764		0.3424

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$$\begin{aligned} b_1 &= \frac{1}{3} \sum y \sin 2x \\ &= \frac{1}{3} [-4.6764] = -1.2124 \end{aligned}$$

$$\begin{aligned} b_2 &= \frac{1}{3} \sum y \sin 4x \\ &= \frac{1}{3} [0.3464] = 0.1155 \end{aligned}$$

Therefore Fourier series is given by

$$f(x) = \frac{a_0}{2} + a_1 \cos 2x + b_1 \sin 2x + a_2 \cos 4x + b_2 \sin 4x$$

$$y = 11.733 - 7.733 \cos 2x - 1.2124 \sin 2x - 2.833 \cos 4x + 0.116 \sin 4x$$

Practical Harmonic Analysis - Problems

2. The following table gives the variations of periodic current over a period T .

t (sec)	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
A (amp)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98

Show that there is a direct current part 0.75 amp in the variable current and obtain the first two harmonics of the Fourier series.

Solution:

The interval of t is $0 \leq t < T$

Here, length of interval is T. That is, $l = \frac{T}{2}$

We observe that the values of A at $t = 0$ and $t = T$ are the same. Hence we omit the last value.

Here **N= 6**

Fourier series up to the second harmonics is given by

$$f(x) = \frac{a_0}{2} + \left(a_1 \cos \frac{\pi x}{l} + b_1 \sin \frac{\pi x}{l}\right) + \left(a_2 \cos \frac{2\pi x}{l} + b_2 \sin \frac{2\pi x}{l}\right)$$

Substituting for $l = \frac{T}{2}$

$$f(x) = \frac{a_0}{2} + \left(a_1 \cos \frac{2\pi x}{T} + b_1 \sin \frac{2\pi x}{T}\right) + \left(a_2 \cos \frac{4\pi x}{T} + b_2 \sin \frac{4\pi x}{T}\right)$$

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$$\begin{aligned}a_0 &= \frac{2}{N} \sum y \\&= \frac{2}{6} \sum y \\&= \frac{1}{3} (4.5) \\a_0 &= \mathbf{1.5}\end{aligned}$$



$$a_n = \frac{2}{N} \sum y \cos\left(\frac{n\pi x}{l}\right)$$

$$= \frac{2}{6} \sum y \cos\left(\frac{n\pi x}{T/2}\right)$$

$$a_n = \frac{1}{3} \sum y \cos\left(\frac{2n\pi x}{T}\right)$$

$$b_n = \frac{2}{N} \sum y \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{2}{6} \sum y \sin\left(\frac{n\pi x}{T/2}\right)$$

$$= \frac{1}{3} \sum y \sin\left(\frac{2n\pi x}{T}\right)$$

Put n=1,2

$$a_1 = \frac{1}{3} \sum y \cos\left(\frac{2\pi x}{T}\right)$$

$$a_2 = \frac{1}{3} \sum y \cos\left(\frac{4\pi x}{T}\right)$$

$$b_1 = \frac{1}{3} \sum y \sin\left(\frac{2\pi x}{T}\right)$$

$$b_2 = \frac{1}{3} \sum y \sin\left(\frac{4\pi x}{T}\right)$$

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$x(t)$	$y(A)$	$\cos\left(\frac{2\pi x}{T}\right)$	$y \cos\left(\frac{2\pi x}{T}\right)$	$\cos\left(\frac{4\pi x}{T}\right)$	$y \cos\left(\frac{4\pi x}{T}\right)$
0	1.98	1	1.98	1	1.98
$T/6$	1.30	0.5	0.65	-0.5	-0.65
$T/3$	1.05	-0.5	-0.525	-0.5	-0.525
$T/2$	1.30	-1	-1.3	1	1.3
$2T/3$	-0.88	-0.5	0.44	-0.5	0.44
$5T/6$	-0.25	0.5	-0.125	-0.5	0.125
Total			1.12		2.67

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$x(t)$	$y(A)$	$\sin\left(\frac{2\pi x}{T}\right)$	$y \sin\left(\frac{2\pi x}{T}\right)$	$\sin\left(\frac{4\pi x}{T}\right)$	$y \sin\left(\frac{4\pi x}{T}\right)$
0	1.98	0	0	0	0
$T/6$	1.30	0.866	1.1258	0.866	1.1258
$T/3$	1.05	0.866	0.9093	-0.866	-0.9093
$T/2$	1.30	0	0	0	0
$2T/3$	-0.88	-0.866	0.76208	0.866	-0.76208
$5T/6$	-0.25	-0.866	0.2165	-0.866	0.2165
Total			3.01368		-0.32908

$$\frac{a_0}{2} = \frac{1.5}{2} = 0.75$$

Therefore, the direct current part = 0.75 amperes

$$a_1 = \frac{1}{3} \sum y \cos \left(\frac{2\pi x}{T} \right) = \frac{1}{3} [1.12] = 0.3733$$

$$b_1 = \frac{1}{3} \sum y \sin \left(\frac{2\pi x}{T} \right) = \frac{1}{3} [3.01368] = 1.00456$$

$$a_2 = \frac{1}{3} \sum y \cos \left(\frac{4\pi x}{T} \right) = \frac{1}{3} [2.67] = 0.89$$

$$b_2 = \frac{1}{3} \sum y \sin \left(\frac{2\pi x}{T} \right) = \frac{1}{3} [-0.32908] = -0.1097$$

Fourier series up to the second harmonics is given by

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \left(a_1 \cos \frac{2\pi x}{T} + b_1 \sin \frac{2\pi x}{T} \right) + \left(a_2 \cos \frac{4\pi x}{T} + b_2 \sin \frac{4\pi x}{T} \right) \\ &= 0.75 + \left(0.3733 \cos \frac{2\pi x}{T} + 1.00456 \sin \frac{2\pi x}{T} \right) + \left(0.89 \cos \frac{4\pi x}{T} - \right. \\ &\quad \left. 0.1097 \sin \frac{4\pi x}{T} \right) \end{aligned}$$



THANK YOU

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